

Framing a Machine Learning Problem - Summary

Definition: Framing a machine learning problem means clearly defining the objective, available data, inputs, outputs, and success criteria before building a model.

Key Points

1. Define the problem and business/research objective.
2. Identify available data and input features; assess data quality.
3. Define the target variable (what the model predicts).
4. Determine the ML task: Classification, Regression, Clustering, Recommendation, or Time Series Forecasting.
5. Select evaluation metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC (classification); MAE, RMSE, R^2 (regression).
6. Consider constraints such as compute resources, privacy, interpretability, deployment, and latency.
7. Typical workflow: Collect data → Explore → Clean & preprocess → Feature engineering → Train/Validation/Test split → Train models → Evaluate → Tune hyperparameters → Deploy & Monitor.

Example

Dog breed identification:

Objective: Predict the breed from an image.

Input: RGB dog image.

Output: One of multiple breed classes.

Problem type: Multi-class classification.

Metrics: Accuracy and Log Loss.